

Sustainable Relationships*

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Abstract

How do firm boundaries shape the allocation of environmentally sustainable inputs in global value chains? In Côte d'Ivoire's cocoa sector, we combine customs records with supply-chain maps linking importers to exporters, cooperatives, departments, and local deforestation. Comparing integrated and non-integrated exporters serving the same importer-year, we find that vertically integrated transactions carry 13% lower cocoa deforestation exposure, about 37 fewer embedded hectares per transaction. The gap is larger for importers with stronger sustainability incentives and smaller for those with more relational sourcing. Our results show that firm boundaries shape how scarce environmentally sustainable inputs are allocated within GVCs.

Keywords: Global value chains; vertical integration; deforestation; traceability; cocoa.

JEL classifications: F14; L14; Q23; Q56.

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1 Introduction

Global value chains (GVCs) are a defining feature of the world economy, linking producers, traders, and consumers across countries. Although participation in GVCs is often viewed as a pathway to growth for low- and middle-income countries, its environmental consequences have become an increasing source of concern (World Bank, 2020). Agricultural GVCs are particularly important in this regard: they account for a substantial share of tropical and subtropical forest loss, making them a major contributor to global environmental degradation (Pendrill et al., 2019; Berman et al., 2023). In response, environmental requirements within GVCs have become increasingly salient. Recent regulatory initiatives, including the European Union Deforestation Regulation (EUDR) and related supply-chain due-diligence rules, reflect growing pressure on international buyers to ensure that their sourcing practices are sustainable.¹

Within these chains, the organization of buyer-supplier relationships is highly heterogeneous. Some lead firms source through vertically integrated relationships, while others primarily rely on non-integrated suppliers (Grossman and Hart, 1986; Lafontaine and Slade, 2007; Alfaro et al., 2025). This heterogeneity is important because organizational form can shape the incentives and control rights that govern sourcing relationships (Macchiavello and Miquel-Florensa, 2018; Hansman et al., 2020). In particular, when low-deforestation inputs are scarce and difficult to contract on, integration may lower deforestation exposure by improving incentives, strengthening control over upstream stages, and limiting diversion of scarce compliant supply. However, the extent to which this occurs in practice is an open question, as the relationship between organizational form and environmental outcomes in GVCs remains poorly understood (Boudreau et al., 2023).

This paper examines how buyer-supplier relationships shape the allocation of environmentally sustainable inputs in GVCs, using the cocoa sector in Côte d’Ivoire. The broader cocoa and chocolate value chain generates more than US\$100 billion annually, chocolate products are consumed worldwide, and cocoa remains a major export crop and source of rural employment in producing countries (Basic and FAO, 2020). These economic benefits coexist with substantial environmental costs. Globally, cocoa is among the leading drivers of commodity-induced deforestation (Global Forest Watch, 2024). In Côte d’Ivoire, the world’s largest cocoa producer, an estimated 2.4 million hectares of forest were converted to cocoa

¹See the European Commission pages on the Regulation on deforestation-free products and its implementation: https://environment.ec.europa.eu/topics/forests/deforestation/regulation-deforestation-free-products_en and https://green-forum.ec.europa.eu/nature-and-biodiversity/deforestation-regulation-implementation_en.

plantations between 2000 and 2019, accounting for roughly 45% of total deforestation and forest degradation in the country over that period (Renier et al., 2023). A key feature of this sector is that cocoa reaching the export gate in Côte d’Ivoire originates from farmers, cooperatives, and departments that differ substantially in deforestation risk and traceability (Guye, 2024). At the same time, the sector relies heavily on vertically integrated relationships where international buyers own domestic exporters (Gayi and Tsowou, 2017).

We study this with detailed data on Ivorian cocoa relationships, combining customs records with granular measures of deforestation and domestic supply-chain links. Our empirical design compares integrated and non-integrated exporter trade flows serving the same importer in the same year.

A simple theoretical framework that adapts Hansman et al. (2020) to environmental compliance organizes the empirical analysis around three testable predictions. As low-deforestation cocoa is hard to contract on, integration should lower deforestation exposure by improving incentives and limiting diversion of scarce compliant supply. The integration gap should be larger when importers have stronger incentives to obtain compliant cocoa, and smaller when relational incentives make delivery by non-integrated exporters more credible.

Consistent with these predictions, integrated exporter transactions exhibit 13% lower deforestation exposure than non-integrated transactions. This estimated gap corresponds to 37 fewer hectares of embedded cocoa-deforestation exposure per transaction.² The integration gap is larger for importers headquartered in the European Union, and it is attenuated for importers with stronger relational sourcing.

Related literature. This paper builds on three strands of literature. First, it contributes to the large literature on trade and the environment, which studies how international trade and GVCs affect environmental outcomes such as deforestation, carbon emissions, and biodiversity loss (Lambin et al., 2018; Abman and Lundberg, 2020; Farrokhi and Lashkaripour, 2021; Hsiao, 2021; Dominguez-Iino, 2026; Carreira et al., 2024; Gianinazzi et al., 2024; Gollin and Wolfersberger, 2025; Noack et al., 2026).³ While this literature has established that trade can be an important driver of environmental degradation, much less is known about how environmental exposure is allocated across firm-to-firm links within a GVC. We address that gap by showing that, within the same importer, integrated and non-integrated exporter links

²In the sample, the average importer–exporter–department–destination–year cell corresponds to about 287 hectares of cocoa deforestation exposure, so the 13% gap is roughly 37 fewer hectares per cell. Integrated links account for about two-thirds of trade volume in our sample, underscoring that the difference is economically meaningful.

³See Copeland et al. (2022) for a review.

differ systematically in the deforestation exposure of the cocoa they supply.

Second, the paper relates to the literature on firm boundaries, vertical integration, and inter-firm relationships (Baker et al., 2001, 2002; Lafontaine and Slade, 2007; Antràs and Chor, 2013; Del Prete and Rungi, 2017; Macchiavello and Miquel-Florensa, 2018; Alfaro et al., 2019; Berlingieri et al., 2021; Alfaro et al., 2025). A central insight of this literature is that organizational form matters particularly when important dimensions of transactions are difficult to observe, specify, or enforce contractually. Recent work shows that integration can improve a hard-to-contract-on quality margin by changing supplier behavior and facilitating quality upgrading (Hansman et al., 2020). We extend this insight to environmental compliance in agricultural GVCs, where the relevant quality attribute is the deforestation-risk and traceability content of cocoa.

Third, the paper speaks to the literature on global, and responsible sourcing. Existing work shows that importers differ in how they organize sourcing, how they allocate purchases across suppliers, and how they respond to supplier heterogeneity (Antras et al., 2017; Bastos et al., 2018; Cajal-Grossi et al., 2023). A related literature emphasizes that governance and repeated interaction can shape labor and social outcomes in supply chains (Macchiavello and Morjaria, 2015; Koenig and Poncet, 2019, 2022; Herkenhoff and Krautheim, 2022; Boudreau, 2024; Alfaro-Ureña et al., 2025; Macchiavello and Morjaria, 2021). We add evidence that importers differ in the environmental attributes of the supply they receive, and that relational sourcing can partially substitute for ownership in shaping those attributes.

The paper is organized as follows. Section 2 provides a background of the Ivorian cocoa sector, describes the data and three stylized facts. Section 3 presents the conceptual framework and the empirical strategy. Section 4 shows the results. Finally, Section 5 concludes.

2 Background, Data, and Facts

2.1 Background

Cocoa production in Côte d’Ivoire begins on smallholder farms, where ripe pods are harvested by hand.⁴ After harvest, beans are fermented, dried, stored in sacks, and sold either to intermediaries, such as *traitants* and *acheteurs*, or to cooperatives, which aggregate cocoa from producers. Cooperative sections operate in most villages, including informal settlements

⁴There are two harvest seasons. The main season runs from October to March, while the mid-crop season runs from April to September.

in former forest areas known as *campements*. Exporters then receive the beans, perform additional controls and commercial preparation, and ship them to international importers.⁵ A large share of Ivorian cocoa exports is contracted before physical delivery.⁶ Industry accounts suggest that roughly 70 percent of the crop is typically sold forward, with the remaining volumes sold during the season on a spot basis.⁷

For our purposes, two features of this supply chain are especially important. First, production often expands through the clearing of forest and bushland rather than through the conversion of existing cropland, making cocoa cultivation a major driver of forest loss in Côte d’Ivoire (World Cocoa Foundation, 2023). Second, traceability is highly uneven. Less than half of total exports can be directly linked to cooperatives, and traceability varies sharply across origin departments (Trase, 2024).

As a result, exporters do not procure a homogeneous input. Instead, they source cocoa that differs substantially in origin, traceability, and implied deforestation risk. Traceability and the physical segregation of cocoa have therefore become central features of procurement. Public company materials make this clear. For example, Cargill highlights farm-to-customer traceability and barcode-based bag tracking, and Barry Callebaut publishes the locations of cooperatives and warehouses in its direct cocoa supply chain.⁸

2.2 Data

Our empirical analysis combines detailed international trade records with high-resolution measures of cocoa-related deforestation and domestic supply-chain links in Côte d’Ivoire. The final dataset is built from three main components, with the aim to trace the environmental attributes embodied in international cocoa shipments.

⁵Cocoa harvested in Côte d’Ivoire can reach export channels within a few weeks: roughly 15 days are required for fermentation and drying, followed by additional time for aggregation, storage, quality control, transport to Abidjan or San Pedro, and shipment. In practice, the lag from harvest to export can range from one to three months. Online Appendix Figures A1 and A2 shows that, three months after the beginning of the harvest season, between 60% and 70% of cocoa has been exported, with no systematic differences in timing between vertically integrated and non-vertically integrated links.

⁶After purchasing an export license for a specific volume of cocoa at a pre-agreed reference price from the Coffee-Cocoa Board (CCC), exporters can sell cocoa to international buyers at a negotiated price. The CCC sets the reference price based on the average of the London and New York futures markets, adjusted for quality and transport costs.

⁷See https://www.lemonde.fr/afrique/article/2024/04/03/cacao-en-cote-d-ivoire-face-a-la-flambee-des-6225780_3212.html.

⁸Official company pages: <https://www.cargill.com/sustainability/community-wellbeing/traceability> and <https://www.barry-callebaut.com/en/sustainability/our-sustainable-raw-materials/transparency-and-traceability-our-cocoa-supply-chain>.

First, we use customs data from Côte d’Ivoire covering the period 2019–2021.⁹ For each transaction, the data report volume in kilograms, CIF value in nominal US dollars, destination, and product type, identified using 6-digit HS codes within heading 18, which covers cocoa and cocoa preparations.¹⁰ Crucially, each record identifies both the importer and the exporter. We standardize company names using a curated dictionary that maps cleaned names to misspellings, alternative spellings, subsidiary names, and corporate groups. This mapping is what allows us to identify whether an importer and an exporter are vertically integrated.¹¹ We found no evidence that any of the importers in our sample completed a M&A during 2019–2021. As a result, the vertical integration indicator is fixed over the sample. We also link importers to their headquarters location and to information on whether they have a sustainability department, using Bureau van Dijk Orbis, company websites and public disclosures.

Second, we combine the international trade data with data on the cocoa cooperatives from which exporters source. This dataset covers more than 4,000 cooperatives across Côte d’Ivoire and includes cooperative names and acronyms, geographic coordinates, buyers identities, and the number of farmers supplying each cooperative (see Online Appendix [Figure A3](#)). The data include updated disclosures by vast majority of large exporters and allow us to distinguish changes in cooperative-exporter links over time ([Guye, 2024](#)). We use these data to map cocoa flows from departments of production to exporters and importers. We first link cooperatives to departments using their geographic coordinates. We then link exporters to cooperatives using company disclosures, and we link importers to exporters using customs data.¹²

Third, to measure cocoa-related deforestation for the years 2019–2021, we use two remote-sensing datasets. The first maps tropical moist forest deforestation and forest degradation annually at 30-meter resolution ([Vancutsem et al., 2021](#)). The second maps cocoa plantations at 10-meter resolution in 2021 ([Kalischek et al., 2023](#)). We overlay these maps to measure cocoa-related deforestation in each department, defined as the area where cocoa plantations overlap with earlier deforestation or forest degradation.

⁹To build alternative measures, we use customs data from Côte d’Ivoire, Ecuador, and Ghana for 2018. Together, these three countries account for roughly 70% of global cocoa production.

¹⁰The main exported cocoa product is cocoa beans, classified under HS code 180100, which represents approximately 50% of international trade transactions in our sample. Following [Trase \(2024\)](#), all empirical results are expressed in bean-equivalent volumes and aggregated across relevant HS codes, excluding HS code 180200, which corresponds to cocoa waste.

¹¹For example, “SACO”, which has been acquired in 2013 by Barry Callebaut, is assigned to the exporter group “Barry Callebaut”.

¹²Although vertical integration between exporters and importers is common (see [Section 2.3](#)), there is no evidence that exporters formally own cooperatives ([Bymolt et al., 2018](#)).

The final dataset contains 58 departments, 77 exporters, 81 importers, 45 destinations and approximately 14,000 observations at the department–exporter–importer–destination–year level for the period 2019–2021.

Measurement. To measure the environmental footprint associated with cocoa trade, we construct cocoa deforestation exposure (CDE), expressed in hectares embedded in traded cocoa. We compute CDE making use of the Trase’s Spatially Explicit Information on Production to Consumption Systems (SEI-PCS) model for Côte d’Ivoire cocoa (Trase, 2024).

First, we consider individual transactions in Ivorian customs declarations for five products: cocoa beans, defatted and non-defatted cocoa paste, cocoa butter, and cocoa powder. We convert the volume of every transaction into cocoa bean-equivalent volume using International Cocoa Organization conversion factors.¹³

Second, we estimate the departmental origin of each trader’s directly sourced cocoa. Direct sourcing refers to cocoa sourced from cooperatives, the most traceable segment of the Ivorian cocoa supply chain. In contrast, indirect sourcing involves intermediate traders. For direct supply chains, we use disclosures of cooperative–trader links. Because these links generally do not report transaction volumes, we convert farmer bases into cocoa volumes using the average annual cocoa production per farmer of 1,225 kg from Bymolt et al. (2018). Missing farmer-base information is extrapolated from other links involving the same cooperative or, if unavailable, from the overall average link size. We then calculate each cooperative’s relative contribution to a trader’s direct farmer base and aggregate these contributions to departments using the cooperatives’ locations, under the assumption that cooperatives source from the departments in which they are located.¹⁴

For each trader, we then calculate the share of volume that is actually directly sourced, and allocate the directly sourced component of each traceable shipment across supplying cooperatives according to their relative contribution to the trader’s direct supply base. These cooperative-level contributions are then summed within departments. For traders with disclosed suppliers, the remaining volume is classified as indirect. For traders without supplier disclosures, and that are not exporting cooperatives, we cannot distinguish direct from indirect sourcing; their shipments are therefore classified as having unknown sourcing.

¹³HS codes are converted into bean equivalents using the following factors: 180100, cocoa beans, has a factor of 1; 180310, cocoa paste, whether or not defatted, not defatted, has a factor of 1.25; 180320, cocoa paste, whether or not defatted, wholly or partly defatted, has a factor of 1.25; 180400, cocoa butter, fat and oil, has a factor of 1.33; and 180500, cocoa powder, not containing added sugar or other sweetening matter, has a factor of 1.18.

¹⁴Renier et al. (2023) show that this is true in most cases, though some of the larger cooperatives which export directly may source from several regions across the country.

Third, we combine departmental sourcing estimates with spatial estimates of cocoa deforestation. Cocoa deforestation is defined as the conversion of undisturbed forest to non-forest at locations that are subsequently mapped as cocoa plantations. In the baseline measure, for each export year, cocoa deforestation is calculated using a four-year lag between deforestation and the first potential cocoa harvest and a 15-year allocation window.¹⁵ Cocoa deforestation exposure is then calculated from the market share of each actor in each department and year. For trade flows that are directly traced to departments, CDE is obtained by assigning departmental cocoa deforestation to traders in proportion to their department-level market shares. Because not all trade flows can be traced to departments, we also allocate cocoa deforestation to indirect and unknown flows. For these untraced flows, we calculate how much each department contributes to national untraced cocoa and uses these departmental weights to assign observed cocoa deforestation to the corresponding trade flows. Summing across departments gives the actor’s total cocoa deforestation exposure.¹⁶

Online Appendix [Table A1](#) reports summary statistics for the main variables used in the analysis.

2.3 Stylized facts

We organize the setting around three stylized facts. Together, these facts imply that the environmental footprint of GVCs depends not only on aggregate export volumes, but also on how origin-specific cocoa is sorted across importer–exporter relationships.

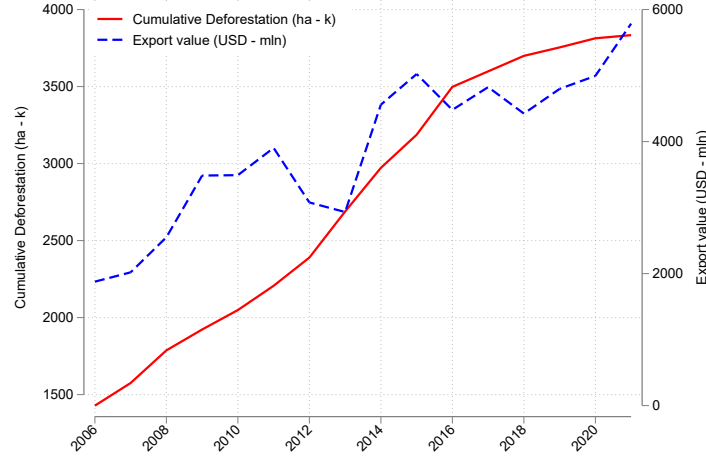
Fact I: Cocoa exports expand during a period of continuing forest loss. The expansion of cocoa exports from Côte d’Ivoire over 2005–2021 took place against a backdrop of sustained deforestation ([Figure 1](#)). This pattern highlights that deforestation-free cocoa is a scarce input. When aggregate exports expand in a setting where deforestation is widespread and persistent, access to cocoa that is credibly unassociated with recent forest loss becomes limited.

This motivates measuring not only how much cocoa is exported, but also where exported cocoa comes from. If cocoa origins differ in their exposure to deforestation, then the environmental content of trade depends on the sub-national geographic composition of exporters’

¹⁵This timing reflects the fact that newly planted cocoa trees typically become productive after about four years. Thus, cocoa exported in 2019 is associated with cocoa planted in 2015 or earlier, and cocoa deforestation for 2019 is measured as deforestation between 2001 and 2015 at cocoa-plantation locations.

¹⁶In Online Appendix [Table A5](#), we show that this assumption is not critical for our results: the main findings are robust to the exclusion of unknown and indirect flows.

Figure 1: Deforestation and cocoa exports



Notes: The figure plots cumulative total deforestation and cocoa export value in Côte d'Ivoire. Deforestation maps are derived from the European Commission Joint Research Centre's Tropical Moist Forest (TMF) dataset, Version 1 (2023), based on 41 years of Landsat time series at 30-meter spatial resolution and aggregated to the department level (Vancutsem et al., 2021). Deforestation refers to the year in which land cover changed from undisturbed or degraded forest to non-forest. Export data are from FAOSTAT and cover cocoa beans, cocoa butter, cocoa fat and oil, cocoa husks and shells, cocoa paste not defatted, and cocoa powder and cake.

sourcing portfolios (see Online Appendix [Figure A4](#)).

Fact II: Importers source from different exporters and departments. Côte d'Ivoire's cocoa importers are exposed to broad geographic sourcing through the exporters from whom they buy. [Table 1](#) documents that the median importer purchases from 8 exporters in a given year. These exporters are themselves geographically diversified: the median exporter sources from 20 departments and serves 6 importers, while the corresponding averages are 23.8 departments and 8.6 importers. As a result, even within a single importer-exporter relationship, cocoa originates from a median of 19 departments.

Table 1: Market Structure: Descriptive Statistics

	Median	Mean	Std. Dev.	Min	Max
Exporters per importer	8.00	9.39	7.32	1	24
Departments per exporter	20.00	23.80	17.84	1	48
Importers per exporter	6.00	8.60	6.82	1	27
Departments per exp.-imp. pair	19.00	20.18	15.14	1	47

Notes: The table reports descriptive statistics for the cocoa export network. The unit of observation is an importer-year for row 1, an exporter-year for rows 2-3, and an exporter-importer pair-year for row 4. Statistics are computed over the pooled sample.

These patterns imply that the same importer can obtain cocoa through several exporters, and each exporter pools cocoa from many departments before selling to multiple importers. This structure creates scope for exporter-specific allocation within the supply chain: importers may differ in their exposure depending on which exporters they buy from and how those exporters allocate cocoa across buyers.

Fact III: Large importers use different sourcing strategies. The import market is highly concentrated, and the largest importers have, at least formally, expressed concern about deforestation. [Table 2](#) documents this. Cargill, Barry Callebaut, and Olam are the three largest importers, and the top 10 together account for 86 percent of the market in our data. The table also shows that most of the largest importers have announced zero-deforestation commitments (ZDCs). Thus, the main importers in this market are not firms for which deforestation is irrelevant; rather, many have publicly committed to reducing or eliminating deforestation from their cocoa supply chains.

At the same time, these importers differ markedly in how they source. Some large importers rely heavily on vertically integrated relationships: for example, Cargill, Barry Callebaut, and Olam source around 70 percent of their volumes within their corporate boundaries. Others, including Ecom and ETC Group, source much smaller shares, while several importers have no vertically integrated sourcing in our data.¹⁷ This means that even among importers with similar deforestation concerns, procurement is not organized in the same way. Some importers pursue low-deforestation cocoa through more integrated supply relationships, while others continue to rely primarily on non-integrated exporters.

¹⁷In Ghana and Ecuador the percentage of vertically integrated transactions is on average lower, 0.24 and 0.12 respectively, but still substantial. This is true in other contexts as well. For instance, several international traders have integrated backward into exporting and processing stages of the coffee chain in sourcing countries ([Del Prete et al., 2026](#)).

Table 2: Top Importers: Descriptive Statistics

	(1)	(2)	(3)	(4)
	Mkt	Num	VI	ZDC
	Share	Destinations	Share	Year
CARGILL	0.19	9.64	0.79	2018
BARRY CALLEBAUT	0.15	23.69	0.70	2015
OLAM	0.13	25.40	0.70	2017
ECOM	0.11	14.73	0.18	2018
SUCDEN	0.10	15.32	0.56	2019
TOUTON	0.10	17.03	0.72	2017
ACT INTERNATIONAL	0.04	15.28	0.00	—
CCT INTERNATIONAL	0.02	9.73	0.00	—
ETC GROUP	0.02	10.30	0.06	2021
AGROFORCE COMMODITIES	0.02	11.97	0.00	—
Top 10	0.86	15.31	0.37	
Top 20	0.98	11.28	0.28	

Notes: The table reports descriptive statistics for the ten largest importers active in Côte d’Ivoire, ranked by average in-sample market share (volumes). Columns (2)–(3) report average values over the pooled sample. VI share is measured in volumes. Column (4) refers to year companies have announced zero-deforestation commitments (ZDCs). The bottom rows report corresponding statistics for the top 10 and top 20 importers. “NA” indicates that the alternative vertical volume share is not applicable; “—” indicates that no ZDC year is observed in the sample.

This matters for our paper for two reasons. First, because a small number of importers account for most cocoa purchases, changes in the sourcing behavior of a few firms can affect a large share of traded cocoa. Importers’ procurement choices are therefore central to understanding how deforestation exposure is allocated across the supply chain. Second, the coexistence of ZDCs and heterogeneous sourcing structures creates the key empirical margin we study. If many importers care about deforestation, but differ in whether they buy through integrated or non-integrated exporters, then differences in deforestation exposure cannot be attributed simply to whether an importer has stated environmental commitments. Instead, they depend on how that commitment is implemented through sourcing relationships.¹⁸

3 Theoretical Framework

In this section we describe how we study the link between vertical integration and deforestation exposure. Our empirical analysis is guided by the theoretical framework on organiza-

¹⁸Online Appendix [Figure A5](#) illustrates this allocation margin in a concrete case. Holding fixed the importer, Barry Callebaut, and the year, 2019, two different exporters are exposed to different bundles of origin departments.

tional structure and output quality developed by [Hansman et al. \(2020\)](#). We first introduce how this framework maps to our setting and then explain our empirical strategy in detail.

Cocoa beans with low deforestation content are an input for which it is difficult to write enforceable contracts.¹⁹ Importers may want to source cleaner beans but cannot fully control which beans an exporter delivers, nor can they easily verify whether an exporter is allocating its most traceable and low-risk supply to them or to other buyers.

The framework in [Hansman et al. \(2020\)](#) formalizes this problem as an incomplete-contracting model with a downstream firm that values input quality and an upstream supplier that controls non-contractible actions before delivery. The supplier can take a quantity-oriented action, a_1 , which raises the value of the input in an outside use, and a quality-oriented action, a_2 , which raises the surplus created for the quality-sensitive buyer. The quality surplus is $Q = Q_0 - \gamma a_1 + \delta a_2$, while the outside-use or quantity surplus is $P = a_1$. Quantity is observable and rewarded in market transactions, but quality is difficult to observe or contract on. When the supplier is independent, it retains control over the input and bargains ex post with the buyer. Because the supplier can always sell to an alternative buyer that values quantity, it has an incentive to raise P even when doing so lowers Q . Vertical integration changes this allocation of control rights: once the downstream firm owns the supplier, the supplier no longer uses the outside option to strengthen its bargaining position, muting the distorted incentive to take quantity-oriented actions that reduce quality.

In our application, the quality surplus Q is the compliance value of cocoa that is traceable and low in deforestation risk. The quantity or outside-use margin P corresponds to the exporter’s ability to sell, or divert cocoa without preserving the low-deforestation attributes valued by a particular importer. Risky or less traceable sourcing lowers compliance quality, while compliance-oriented effort raises it. Deforestation exposure is therefore decreasing in this relationship-specific quality. Integration gives the importer stronger control over delivery and over the allocation of scarce traceable beans, while a non-integrated exporter retains control over the cocoa and can allocate low-risk supply to alternative buyers. The dynamic extension in [Hansman et al. \(2020\)](#), which closely follows [Baker et al. \(2001, 2002\)](#), also clarifies why repeated relationships may partially substitute for ownership: when the continuation value of the relationship is high enough, relational contracts can support quality-oriented actions even without integration, although outside-use incentives make such arrangements harder to

¹⁹The Federation of Cocoa Commerce (FCC) publishes standardized Short Form Contracts, often referred to as SFC or CP3/CP4 contracts, designed to streamline physical cocoa international trade. These contracts incorporate the comprehensive FCC Contract Rules, including quality requirements such as bean size and moisture content, but they rarely include origin requirements. See <https://www.cocoafederation.com/services/rules/short-form-contracts>.

sustain.

Adapting [Hansman et al. \(2020\)](#) to our setting yields three testable predictions: (i) integrated links deliver lower deforestation exposure, (ii) stricter downstream compliance widens the integration gap, and (iii) continuation value narrows the integration gap. In what follows, we discuss each prediction and its empirical implementation in turn.

In all three predictions, the dependent variable, D_{iepdt} , is the log of cocoa deforestation exposure in hectares embodied in shipments from exporter e to importer i , from origin department p , to destination country d , in year t .²⁰ Throughout, identification comes from comparing integrated and non-integrated exporter flows serving the same importer in the same year, including importer-year fixed effects (μ_{it}).

Prediction 1: Integrated links deliver lower deforestation exposure. The first prediction follows from the static hold-up problem in [Hansman et al. \(2020\)](#): within an importer-year, shipments from vertically integrated exporters should have lower deforestation exposure than shipments from non-integrated exporters.

We estimate:

$$D_{iepdt} = \alpha + \beta VI_{ie} + \mathbf{X}'_{iepdt} \delta + \mu_{it} + \varepsilon_{iepdt}, \quad (1)$$

where VI_{ie} equals one if importer i and exporter e belong to the same corporate group. VI_{ie} is fixed at the importer-exporter level over the sample. $\mathbf{X}_{iepdt}$ includes link-level trade controls such as shipment volume and price (unit values). Since we argue that vertical integration enables importers to have access to cocoa beans with a lower deforestation exposure we expect to see $\beta < 0$.

Prediction 2: Stricter downstream compliance widens the integration gap. If the deforestation-content works as quality, then we should also expect to see the gap to be larger when the importer places a higher value on low-deforestation cocoa, for instance because it faces stronger regulatory or reputational pressure downstream, since stronger compliance incentives increase the effort that integrated exporters can implement but non-integrated exporters cannot.

To test this, we interact vertical integration with importer-level proxies for downstream

²⁰Because deforestation exposure in hectares is strictly positive in our sample, although sometimes very small ([Table 1](#)), the log transformation does not drop any observations.

compliance pressure:

$$D_{iepdt} = \alpha + \beta VI_{ie} + \lambda (VI_{ie} \times Stringency_i) + \mathbf{X}'_{iepdt} \delta + \mu_{it} + \varepsilon_{iepdt}. \quad (2)$$

Because μ_{it} absorbs the level effect of $Stringency_i$, only the interaction is identified. We use European Union headquarters and the presence of a sustainability department as proxies for a higher marginal value of low-deforestation supply.²¹ Given that we argue that higher compliance requirements make the integration more valuable we expect to see $\lambda < 0$.

Prediction 3: Continuation value narrows the integration gap. Finally, as in the dynamic extension of [Hansman et al. \(2020\)](#), repeated interactions can partially substitute for ownership. When an exporter values its relationship with an importer, the prospect of losing future business makes diversion of compliant supply more costly, supporting credible delivery without formal ownership. The framework accordingly predicts that the integrated-link advantage is smaller when the importer is more "relational": a higher continuation value relaxes the non-integrated exporter's incentive to divert, narrowing but not necessarily eliminating the gap.

To test whether continuation value attenuates the non-integrated disadvantage, we estimate

$$D_{iepdt} = \alpha + \beta VI_{ie} + \gamma (VI_{ie} \times ContValue_i) + \mathbf{X}'_{iepdt} \delta + \mu_{it} + \varepsilon_{iepdt}, \quad (3)$$

where $ContValue_i$ captures importer-level characteristics that increase the value, for an exporter, of maintaining the relationship with importer i . Our primary proxy is importer relationality, described in [Online Section A.3](#): a predetermined, standardized measure constructed from 2018 customs data following [Cajal-Grossi et al. \(2023\)](#), where higher values indicate relational sourcing.

Because importer-year fixed effects absorb the level effect of $ContValue_i$, identification comes from whether the vertically integrated link advantage varies systematically with importer-level continuation value. The fact that relational contract may decrease the value from integration we expect to see: $\beta < 0$ and $\gamma > 0$. The integrated-link advantage remains negative but becomes smaller when continuation value is higher.

²¹We prefer to use these proxies, instead of destination markets, as countries of destination recorded in customs data may be a processing/trading hub rather than the final consumer market.

4 Results

The results in [Table 3](#) show a negative association between vertical integration and deforestation exposure. Across all specifications, the coefficient on VI is negative, highly statistically significant, and stable. The coefficient in column (3) implies that integrated links have approximately 13% lower deforestation exposure than non-integrated links serving the same importer-year.²²

The framework points to two complementary mechanisms. First, integration may increase the stock of low-risk cocoa generated ex ante by making compliance-oriented effort easier to implement. Second, integration may affect the relationship-specific allocation and delivery of scarce traceable cocoa ex post, because non-integrated exporters retain stronger outside-use incentives.

Table 3: Deforestation and Vertical Integration

Dep. Var.	Deforestation		
	(1)	(2)	(3)
VI	-0.139*** (0.020)	-0.138*** (0.020)	-0.140*** (0.019)
Obs.	14,067	14,067	14,067
FEs	i, y	i, y	iy
Volume	✓	✓	✓
Price		✓	✓
R^2	0.96	0.96	0.96
Mean DV	3.18	3.18	3.18

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific measure of vertical integration, defined as an indicator equal to one when the importer and exporter belong to the same corporate group. Controls include log export volume and log price as indicated. Fixed effects are reported in the row labeled FEs. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Online [Table A2](#) speaks to the exporter-level ex ante component. At the exporter-department-year level, integrated exporters are associated with lower deforestation exposure. Since deforestation exposure is decreasing in compliance quality, this pattern is consistent with integrated exporters preserving cocoa with higher compliance quality, through lower risky sourcing, higher compliance, or both. This table therefore supports an ex ante sourcing or compliance-quality margin, but it does not by itself establish that the relevant quality object is relationship-specific.

²²The semi-elasticity is $100 \times (e^{-0.140} - 1) \simeq -13.1\%$. All coefficients reported in the text are calculated with the same transformation.

Online [Table A3](#) speaks more directly to the relationship-level compliance quality. It compares three types of importer–exporter flows: the importer’s own integrated exporter, stand-alone exporters, and subsidiaries of other importers. Stand-alone and outside subsidiaries’ flows both have significantly higher deforestation exposure than the importer’s own integrated flows. If the integration advantage were driven only by exporter-level ex ante effort differences, deliveries from subsidiaries of other importers would be expected to look more like the importer’s own integrated exporter. Instead, they resemble stand-alone exporters. This pattern suggests that ownership matters for the specific importer–exporter relationship: cleaner, more traceable cocoa is more likely to be delivered to the parent importer. The evidence is therefore consistent with the relationship-level compliance quality, and with an ex post allocation or delivery margin in addition to any exporter-level sourcing differences. Importantly, this exercise does not by itself prove that integration reduces aggregate deforestation; rather, it helps separate the within-chain allocation mechanism from a broader production or environmental-improvement mechanism. This distinction is central to our interpretation: vertical integration shapes the environmental content embodied in a given importer–exporter link, but the aggregate welfare implications depend on whether the observed reallocation is accompanied by additional investments that expand the overall supply of low-deforestation cocoa.

We then examine whether the relationship between vertical integration and deforestation exposure varies with importer characteristics, focusing on Predictions 2 and 3 ([Table 4](#)).

In column (1) the coefficient on $VI \times HQ\ EU$ is -0.150 , implying that the deforestation-reducing association of vertical integration is stronger for importers headquartered in the European Union. The implied effect for EU-headquartered importers is approximately -17.6% . This result is consistent with Prediction 2: vertical integration is more strongly associated with lower deforestation exposure when importers have stronger institutional incentives to monitor sustainability outcomes.

In column (2), the coefficient on VI is negative and significant, while the interaction between VI and importer relationality is positive and significant. Since relationality is a predetermined importer-level measure of global relational sourcing, the positive interaction implies that the negative association between vertical integration and deforestation exposure becomes weaker for importers that rely more heavily on relational sourcing. Consistent with Prediction 3, when importers already have stronger relational ties, the additional role of vertical integration in reducing deforestation exposure appears less pronounced.

Table 4: Deforestation and Vertical Integration

Dep. Var.	Deforestation	
	Stringency (1)	ContValue (2)
VI	-0.044 (0.045)	-1.024*** (0.150)
$VI \times HQ\ EU$	-0.150*** (0.047)	
$VI \times Relationality$		0.513*** (0.085)
Obs.	12,804	13,353
FEs	iy	iy
Controls	✓	✓
R^2	0.96	0.96
Mean DV	3.27	3.22

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressors are the baseline vertical-integration indicator and its interaction with importer characteristics. The vertical-integration indicator equals one when the importer and exporter belong to the same corporate group. HQ EU equals one when the importer’s headquarters are located in the European Union. Relationality is a standardized importer-level metric described in Online [Section A.3](#). Controls include log export volume and log price as indicated. All specifications include importer–year fixed effects. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

4.1 Robustness checks

The main findings hold under a variety of alternative specifications, reported in the Online Appendix. Online Appendix [Table A4](#) shows that the estimates are unchanged when we use an alternative clustering scheme for inference, namely two-way clustering. The results also hold under different sample restrictions, including specifications that exclude unknown flows, only consider direct flows, exclude or retain the top 10 importers, and exclude importers that only have one exporter or fewer than three exporters per year (Online Appendix [Table A5](#)). The findings also hold when deforestation exposure is measured per tonne, indicating that the relationship is not mechanically driven by larger shipments carrying more embedded deforestation (Online Appendix [Table A6](#)).

To address concerns about omitted origin-specific factors, we add department-level controls for nighttime lights, the number of cooperatives, mining activity, and land use; the coefficient on vertical integration is robust to these additions (Online Appendix [Table A7](#)). We also implement the selection diagnostics of [Altonji et al. \(2005\)](#) and [Oster \(2019\)](#). Online Appendix [Table A8](#) reports the baseline coefficient on the vertical integration indicator, and

then bias-adjusted coefficients under increasing values of the relative selection parameter δ , which captures the strength of selection on unobservables relative to selection on observables. The coefficient remains negative across all specifications, and the value of δ required to reduce the estimated effect to zero is 1.486, a large positive value indicating that unobservables would need to be implausibly strong to eliminate the vertical integration advantage.

The vertical integration coefficient remains negative and significant even when including exporter fixed effects, indicating that the advantage is not only driven by exporter-level selection into cleaner networks. Online Appendix [Table A9](#) includes even more demanding fixed effects, such as exporter-year and destination-year fixed effects, and shows that the coefficient becomes smaller, but remains negative and significant across all specifications. Together, these results are consistent with Prediction 1: when importers source from vertically integrated exporters they can obtain cocoa with lower deforestation exposure, even after accounting for exporter-specific characteristics.

Moreover, Online Appendix [Table A10](#) provides evidence consistent with the allocation channel by examining whether vertically integrated links are also associated with higher shares of direct and certified sourcing. The result supports the interpretation that firm boundaries affect the allocation of traceable, certified cocoa across downstream buyers.

Finally, Online Appendix [Table A11](#) shows that Predictions 2 and 3 hold when using alternative proxies for stringency and continuation value. The implied effect of vertical integration for importers with a sustainability department is approximately -13.1%. The coefficient on *VI* is also negative and statistically significant when a relationality dummy is used as an alternative continuation-value proxy, while the interaction with relationality is positive and significant.

5 Conclusions

This paper documents a relationship between organizational forms and the deforestation embodied in cocoa shipments from Côte d’Ivoire. Using importer-by-year comparisons, we show that vertically integrated links are associated with lower cocoa deforestation exposure than non-integrated links serving the same importer.

The main contribution is to show that environmental performance in agricultural GVCs depends not only on which suppliers are used, but also on how heterogeneous inputs are allocated across buyers. When compliant, traceable cocoa is scarce, exporters have an incentive to allocate cleaner beans to buyers whose relationships or ownership links make compliance

most valuable. Vertical integration therefore matters not only because it may raise average sourcing standards, but because it changes how already heterogeneous and scarce sustainable inputs are sorted within the chain.

This has important implications for traceability-based regulation such as the EUDR, as stronger traceability demand may primarily reallocate scarce compliant cocoa toward buyers with stronger relationships, rather than immediately reducing aggregate forest loss. The policy takeaway is that regulations based on documented origin and traceability should be designed with attention to within-chain sorting. If traceable cocoa remains scarce, regulations may shift environmental risk from some traders to others without guaranteeing a proportional reduction in total deforestation. To secure broader environmental benefits, policy should also encourage the expansion of traceable, low-deforestation supply and the upstream investments that make such supply less scarce.

More broadly, the paper highlights that understanding the environmental footprint of GVCs requires examining how heterogeneous inputs are sorted across buyers and sellers, and whether observed reallocation is accompanied by an expansion of the sustainable input supply.

References

- Abman, R. and C. Lundberg (2020). Does free trade increase deforestation? the effects of regional trade agreements. *Journal of the Association of Environmental and Resource Economists* 7(1), 35–72.
- Alfaro, L., D. Chor, P. Antras, and P. Conconi (2019). Internalizing global value chains: A firm-level analysis. *Journal of Political Economy* 127(2), 508–559.
- Alfaro, L., P. Conconi, F. Kamal, and Z. Kroff (2025). Trade within multinational boundaries. Technical report, National Bureau of Economic Research.
- Alfaro-Ureña, A., B. Faber, C. Gaubert, I. Manelici, and J. P. Vasquez (2025). Responsible sourcing? evidence from costa rica.
- Altonji, J. G., T. E. Elder, and C. R. Taber (2005). Selection on observed and unobserved variables: Assessing the effectiveness of catholic schools. *Journal of Political Economy* 113(1), 151–184.
- Antràs, P. and D. Chor (2013). Organizing the global value chain. *Econometrica* 81(6), 2127–2204.

- Antras, P., T. C. Fort, and F. Tintelnot (2017). The margins of global sourcing: Theory and evidence from us firms. *American Economic Review* 107(9), 2514–2564.
- Baker, G., R. Gibbons, and K. J. Murphy (2001). Bringing the market inside the firm? *American Economic Review* 91(2), 212–218.
- Baker, G., R. Gibbons, and K. J. Murphy (2002). Relational contracts and the theory of the firm. *The Quarterly Journal of Economics* 117(1), 39–84.
- Basic and FAO (2020). Comparative study on the distribution of value in european chocolate chains.
- Bastos, P., J. Silva, and E. Verhoogen (2018). Export destinations and input prices. *American Economic Review* 108(2), 353–392.
- Berlingieri, G., F. Pisch, and C. Steinwender (2021). Organizing global supply chains: input-output linkages and vertical integration. *Journal of the European Economic Association* 19(3), 1816–1852.
- Berman, N., M. Couttenier, A. Leblois, and R. Soubeyran (2023). Crop prices and deforestation in the tropics. *Journal of Environmental Economics and Management* 119, 102819.
- Boudreau, L. (2024). Multinational enforcement of labor law: Experimental evidence on strengthening occupational safety and health committees. *Econometrica* 92(4), 1269–1308.
- Boudreau, L., J. Cajal-Grossi, and R. Macchiavello (2023). Global value chains in developing countries: a relational perspective from coffee and garments. *Journal of Economic Perspectives* 37(3), 59–86.
- Bymolt, R., A. Laven, and M. Tyszler (2018). Demystifying the cocoa sector in ghana and côte d’ivoire.
- Cajal-Grossi, J., D. Del Prete, and R. Macchiavello (2023). Supply chain disruptions and sourcing strategies. *International Journal of Industrial Organization* 90, 103004.
- Cajal-Grossi, J., R. Macchiavello, and G. Noguera (2023). Buyers’ sourcing strategies and suppliers’ markups in bangladeshi garments. *The Quarterly Journal of Economics* 138(4), 2391–2450.

- Carreira, I., F. Costa, and J. P. Pessoa (2024). The deforestation effects of trade and agricultural productivity in brazil. *Journal of Development Economics* 167, 103217.
- Copeland, B., S. Shapiro, and M. Scott Taylor (2022). Globalization and the environment. *Handbook of International Economics* 5, 61–146.
- Del Prete, D., P. Fajgelbaum, A. Khandelwal, and R. Macchiavello (2026). Market power along the coffee global value chain. Unpublished.
- Del Prete, D. and A. Rungi (2017). Organizing the global value chain: A firm-level test. *Journal of International Economics* 109, 16–30.
- Dominguez-Iino, T. (2026). Efficiency and redistribution in environmental policy: An equilibrium analysis of agricultural supply chains. *Journal of Political Economy* 134(4), 1075–1109.
- Farrokhi, F. and A. Lashkaripour (2021). Can trade policy mitigate climate change. *Unpublished Working Paper*.
- Gayi, S. K. and K. Tsowou (2017). *Cocoa industry: Integrating small farmers into the global value chain*. United Nations.
- Gianinazzi, V., V. Girard, M. Lehlali, and M. P. Prado (2024). Socially responsible investing and multinationals’ pollution: Evidence from global remote sensing data. *SSRN*.
- Global Forest Watch (2024). Deforestation linked to agriculture indicator. *Washington, DC: World Resources Institute*.
- Gollin, D. and J. Wolfersberger (2025). Agricultural trade and deforestation: the role of new roads. *The Economic Journal*, ueaf125.
- Grossman, S. J. and O. D. Hart (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. *Journal of Political Economy* 94(4), 691–719.
- Guye, V. (2024). Côte d’Ivoire cocoa exports and deforestation. *Trase*.
- Hansman, C., J. Hjort, G. Leon-Ciliotta, and M. Teachout (2020). Vertical integration, supplier behavior, and quality upgrading among exporters. *Journal of Political Economy* 128(9), 3570–3625.
- Heise, S., J. R. Pierce, G. Schaur, and P. K. Schott (2024). Tariff rate uncertainty and the structure of supply chains. Technical report, National Bureau of Economic Research.

- Herkenhoff, P. and S. Krautheim (2022). The international organization of production in the regulatory void. *Journal of International Economics* 137, 103572.
- Hsiao, A. (2021). Coordination and commitment in international climate action: evidence from palm oil. *Unpublished, Department of Economics, MIT*.
- Kalischek, N., N. Lang, C. Renier, R. C. Daudt, T. Addoah, W. Thompson, W. J. Blaser-Hart, R. Garrett, K. Schindler, and J. D. Wegner (2023). Cocoa plantations are associated with deforestation in côte d’ivoire and ghana. *Nature Food* 4(5), 384–393.
- Koenig, P. and S. Poncet (2019). Social responsibility scandals and trade. *World Development* 124, 104640.
- Koenig, P. and S. Poncet (2022). The effects of the rana plaza collapse on the sourcing choices of french importers. *Journal of International Economics* 137, 103576.
- Lafontaine, F. and M. Slade (2007). Vertical integration and firm boundaries: The evidence. *Journal of Economic literature* 45(3), 629–685.
- Lambin, E. F., H. K. Gibbs, R. Heilmayr, K. M. Carlson, L. C. Fleck, R. D. Garrett, Y. Le Polain de Waroux, C. L. McDermott, D. McLaughlin, P. Newton, et al. (2018). The role of supply-chain initiatives in reducing deforestation. *Nature Climate Change* 8(2), 109–116.
- Macchiavello, R. (2022). Relational contracts and development. *Annual Review of Economics* 14(1), 337–362.
- Macchiavello, R. and J. Miquel-Florensa (2018). Vertical integration and inter-firm relationships in the costa rica coffee chain. Working paper.
- Macchiavello, R. and A. Morjaria (2015). The value of relationships: Evidence from a supply shock to kenyan rose exports. *American Economic Review* 105(9), 2911–2945.
- Macchiavello, R. and A. Morjaria (2021). Competition and relational contracts in the rwanda coffee chain. *The Quarterly Journal of Economics* 136(2), 1089–1143.
- Noack, F., D. Rohner, and T. Sonno (2026). The environmental impact of multinational firms. *Nature Climate Change*.
- Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics* 37(2), 187–204.

- Padilla, A., D. Otarod, S. Deloach-Overton, R. Kemna, P. Freeman, E. Wolfe, L. Bird, A. Gulley, M. Trippi, C. Dicken, et al. (2021). Compilation of geospatial data (gis) for the mineral industries and related infrastructure of africa. *US Geological Survey Data Release*.
- Pendrill, F., U. M. Persson, J. Godar, and T. Kastner (2019). Deforestation displaced: trade in forest-risk commodities and the prospects for a global forest transition. *Environmental Research Letters* 14(5), 055003.
- Renier, C., M. Vandromme, P. Meyfroidt, V. Ribeiro, N. Kalischek, and E. K. Zu Ermgassen (2023). Transparency, traceability and deforestation in the ivorian cocoa supply chain. *Environmental Research Letters* 18(2), 024030.
- Trase (2024). Sei-pcs côte d’ivoire cocoa v1.1 supply chain map: Data sources and methods.
- Vancutsem, C., F. Achard, J.-F. Pekel, G. Vieilledent, S. Carboni, D. Simonetti, J. Gallego, L. E. Aragao, and R. Nasi (2021). Long-term (1990–2019) monitoring of forest cover changes in the humid tropics. *Science advances* 7(10), eabe1603.
- World Bank (2020). World development report 2020: Trading for development in the age of global value chains. *DC: World Bank*.
- World Cocoa Foundation (2023). Journey of chocolate: From bean to bar.

A Online Appendix

A.1 Tables

Table A1: Summary statistics

	Mean	Median	SD	Min	Max
Exporters	68.579	69.000	1.274	67.000	70.000
Importers	93.940	90.000	7.875	86.000	104.000
Destinations	42.000	42.000	0.000	42.000	42.000
Deforestation	287.342	29.832	1266.467	0.000	74487.734
VI	0.679	1.000	0.467	0.000	1.000

Notes: The table reports summary statistics for the variables used in the empirical analysis. The sample covers cocoa exports from Côte d'Ivoire over 2019–2021. Exporters, importers, and countries of destination are counted by year. Deforestation denotes cocoa deforestation exposure in hectares under the baseline 15-year allocation window; note that deforestation is never a true zero in our sample as the min is .0002. The vertical-integration variable is an indicator equal to one when the importer and exporter belong to the same corporate group.

Table A2: Deforestation and Vertical Integration - Ex ante effort

Dep. Var.	Deforestation		
	(1)	(2)	(3)
<i>VI</i>	-0.183*** (0.031)	-0.177*** (0.031)	-0.186* (0.094)
Obs.	819	819	819
FEs	y	y	e, y
Volume	✓	✓	✓
Price		✓	✓
R^2	0.87	0.87	0.88
Mean DV	7.35	7.35	7.35

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an exporter–department–year cell. Controls include log export volume and log price as indicated. Fixed effects are reported in the row labeled FEs. Standard errors clustered at the exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A3: Deforestation and Vertical Integration - Ex post delivery

Dep. Var.	Deforestation		
	(1)	(2)	(3)
<i>Stand Alone</i>	0.123*** (0.027)	0.123*** (0.027)	0.123*** (0.027)
<i>Subsidiary</i>	0.159*** (0.021)	0.158*** (0.021)	0.161*** (0.020)
Obs.	14,067	14,067	14,067
FEs	i, y	i, y	iy
Volume	✓	✓	✓
Price		✓	✓
R^2	0.96	0.96	0.96
Mean DV	3.18	3.18	3.18

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The omitted category is a vertically integrated importer–exporter link, defined as a transaction in which the exporter belongs to the importer’s corporate group. Stand Alone identifies non-integrated exporters. Subsidiary identifies exporters integrated with another importer but not with the importer in the observed transaction. Controls include log export volume and log price as indicated. Fixed effects are reported in the row labeled FEs. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A4: Deforestation and Vertical Integration - Alternative clustering

Dep. Var.	Deforestation		
	(1)	(2)	(3)
<i>VI</i>	-0.139*** (0.030)	-0.138*** (0.030)	-0.140*** (0.029)
Obs.	14,067	14,067	14,067
FEs	i, y	i, y	iy
Volume	✓	✓	✓
Price		✓	✓
R^2	0.96	0.96	0.96
Mean DV	3.18	3.18	3.18

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific vertical-integration indicator, equal to one when the importer and exporter belong to the same corporate group. Controls include log export volume and log price as indicated. Fixed effects are reported in the row labeled FEs. The abbreviations are importer i, year y, and importer-year iy. Relative to the baseline table, this table reports inference using an alternative clustering level. Two-way clustered standard errors, at the importer and exporter are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A5: Deforestation and Vertical Integration - Sample composition

Dep. Var.	Deforestation					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>VI</i>	-0.141*** (0.020)	-0.127*** (0.037)	-0.136*** (0.020)	-0.169** (0.077)	-0.142*** (0.020)	-0.141*** (0.020)
Obs.	13,297	12,079	10,877	3,190	12,523	11,983
FEs	iy	iy	iy	iy	iy	iy
Controls	✓	✓	✓	✓	✓	✓
Sample	Direct & Indirect	Direct	Top10	No Top10	ExI>1	ExI>3
R^2	0.95	0.94	0.95	0.98	0.96	0.96
Mean DV	3.02	2.70	3.19	3.14	3.35	3.34

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific vertical-integration indicator, equal to one when the importer and exporter belong to the same corporate group. Controls include log export volume and log price as indicated. All specifications include importer–year fixed effects (iy). The sample used in each column is reported in the row labeled Sample. ExI stands for unique exporter per importer link in a year. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A6: Deforestation and Vertical Integration - Per tonne

Dep. Var.	Deforestation		
	(1)	(2)	(3)
<i>VI</i>	-0.183*** (0.018)	-0.183*** (0.018)	-0.185*** (0.018)
Obs.	14,067	14,067	14,067
FEs	i, y	i, y	iy
Price		✓	✓
R^2	0.05	0.05	0.05
Mean DV	-0.50	-0.50	-0.50

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha) per tonne of cocoa exported. The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific vertical-integration indicator, equal to one when the importer and exporter belong to the same corporate group. Controls include log price as indicated. Fixed effects are reported in the row labeled FEs. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A7: Deforestation and Vertical Integration - Department Controls

Dep. Var.	Deforestation				
	(1)	(2)	(3)	(4)	(5)
<i>VI</i>	-0.140*** (0.037)	-0.189*** (0.044)	-0.129*** (0.035)	-0.168*** (0.034)	-0.229*** (0.044)
Obs.	12,079	12,079	12,079	12,079	12,079
FEs	iy	iy	iy	iy	iy
Controls	✓	✓	✓	✓	✓
Nighttime Lights	✓				✓
Num Coops		✓			✓
Mining			✓		✓
Land use				✓	✓
R^2	0.95	0.94	0.95	0.95	0.96
Mean DV	2.70	2.70	2.70	2.70	2.70

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific vertical-integration indicator, equal to one when the importer and exporter belong to the same corporate group. Controls include log export volume and log price as indicated. Additional department-level controls are reported by column and include nighttime lights, the number of cooperatives, mining activity, and land-use exposure. All specifications include importer–year fixed effects. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A8: Test for Selection on Unobservables

Main result	$\delta = 0.1$	$\delta = 0.2$	$\delta = 0.4$	$\delta = 0.6$	$\delta = 0.8$	$\delta = 1$	δ for $\beta = 0$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>VI</i>	-0.140	-0.131	-0.123	-0.105	-0.087	-0.069	-0.049
							1.486

Notes: The table reports coefficient-stability diagnostics following [Oster \(2019\)](#). Column (1) reports the main estimate. Columns (2)–(7) report bias-adjusted coefficients under alternative values of the relative selection parameter δ . Column (8) reports the value of δ that would be required to reduce the estimated coefficient on vertical integration to zero.

Table A9: Deforestation and Vertical Integration - Alternative FEs

Dep. Var.	Deforestation		
	(1)	(2)	(3)
<i>VI</i>	-0.059** (0.027)	-0.065*** (0.025)	-0.138*** (0.053)
Obs.	14,058	14,036	13,779
FEs	i, e, d, y	iy, dy, ey	idy, ey
Controls	✓	✓	✓
R^2	0.96	0.96	0.96
Mean DV	3.18	3.17	3.12

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific vertical-integration indicator, equal to one when the importer and exporter belong to the same corporate group. Controls include log export volume and log price as indicated. Fixed effects are reported in the row labeled FEs and vary across columns. The abbreviations are importer i, exporter e, destination d, year y, importer-year iy, destination-year dy, exporter-year ey, and importer-department-year idy. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table A10: Traceability and Vertical Integration

Dep. Var.	Share Direct (1)	Share Certified (2)
<i>VI</i>	0.372*** (0.058)	0.284*** (0.031)
Obs.	14,067	14,067
FEs	iy	iy
Controls	✓	✓
R^2	0.74	0.73
Mean DV	0.64	0.40

Notes: The table shows regressions of traceability outcomes on vertical integration. The dependent variable is the share of direct sourcing in column (1) and the share of certified sourcing in column (2). The unit of observation is an importer–exporter–destination–department–year cell. The main regressor is the baseline, relationship-specific vertical-integration indicator, equal to one when the importer and exporter belong to the same corporate group. Controls include log export volume and log price as indicated. All specifications include importer–year fixed effects. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

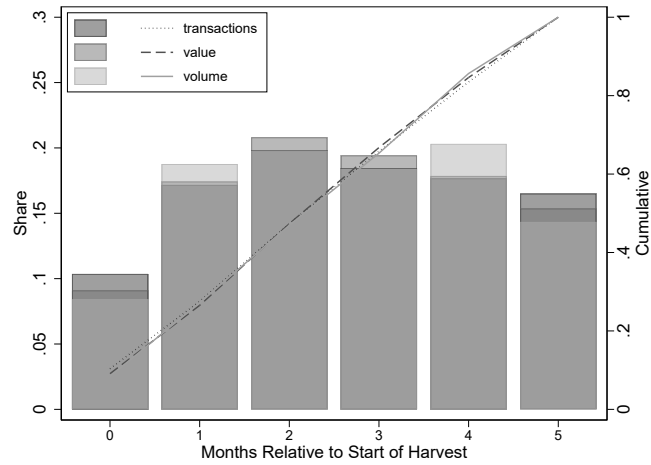
Table A11: Deforestation and Vertical Integration - Alternative proxies

Dep. Var.	Deforestation	
	Stringency (1)	ContValue (2)
<i>VI</i>	0.024 (0.055)	-0.350*** (0.033)
<i>VI × Sustainability Dep</i>	-0.165*** (0.058)	
<i>VI × Relationality</i>		0.221*** (0.037)
Obs.	13,275	13,353
FEs	iy	iy
Controls	✓	✓
R^2	0.96	0.96
Mean DV	3.23	3.22

Notes: The table shows regressions of the log of cocoa deforestation exposure in hectares (ha). The unit of observation is an importer–exporter–destination–department–year cell. The main regressors are the baseline vertical-integration indicator and its interaction with importer characteristics. The vertical-integration indicator equals one when the importer and exporter belong to the same corporate group. Relationality is an indicator that takes value one for buyers above the 90th percentile of the distribution of the (continuous) corresponding sourcing metric. Controls include log export volume and log price as indicated. All specifications include importer–year fixed effects. Standard errors clustered at the importer–exporter level are shown in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

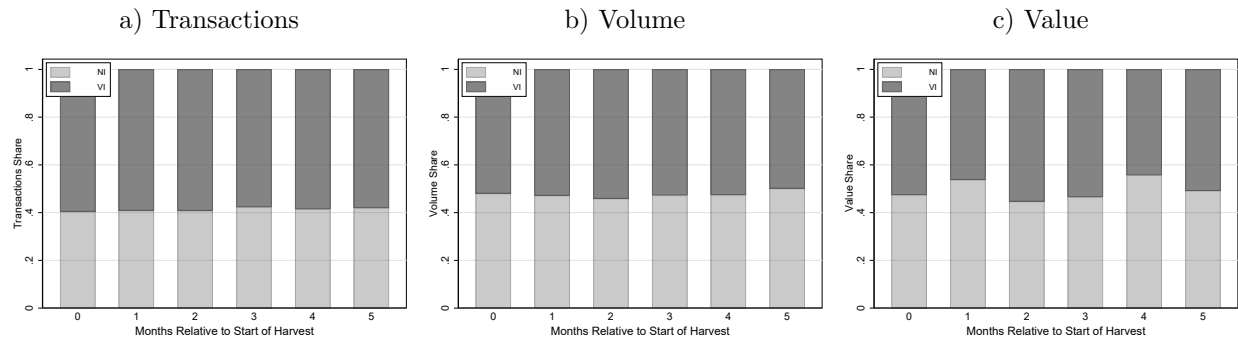
A.2 Figures

Figure A1: Transactions Timing



Notes: The horizontal axis is the number of months from the beginning of the harvest season (at zero).

Figure A2: Transactions Timing by vertical integration



Notes: This figure shows the non-vertically (no-VI) and vertically integrated (VI) shares across the number of months from the beginning of the harvest season (at zero).

Figure A3: Cocoa Cooperative Location

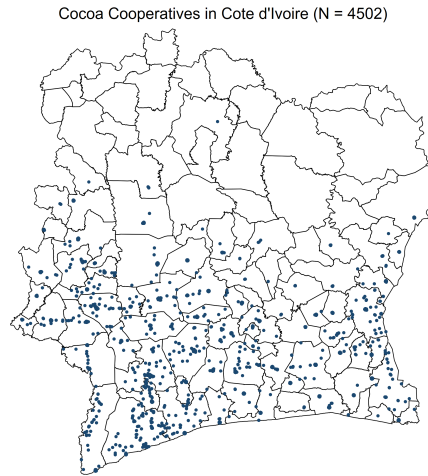
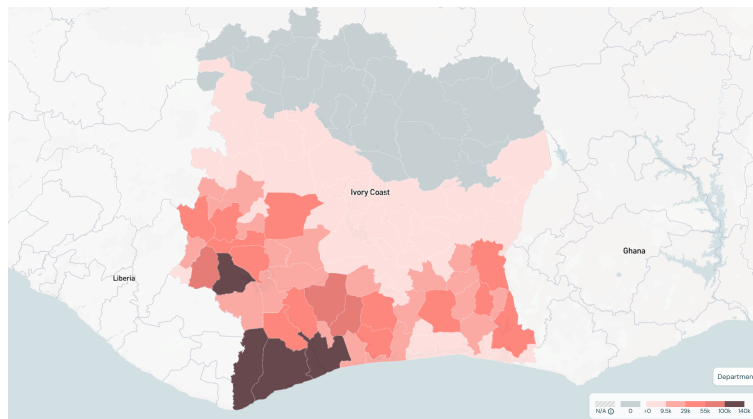
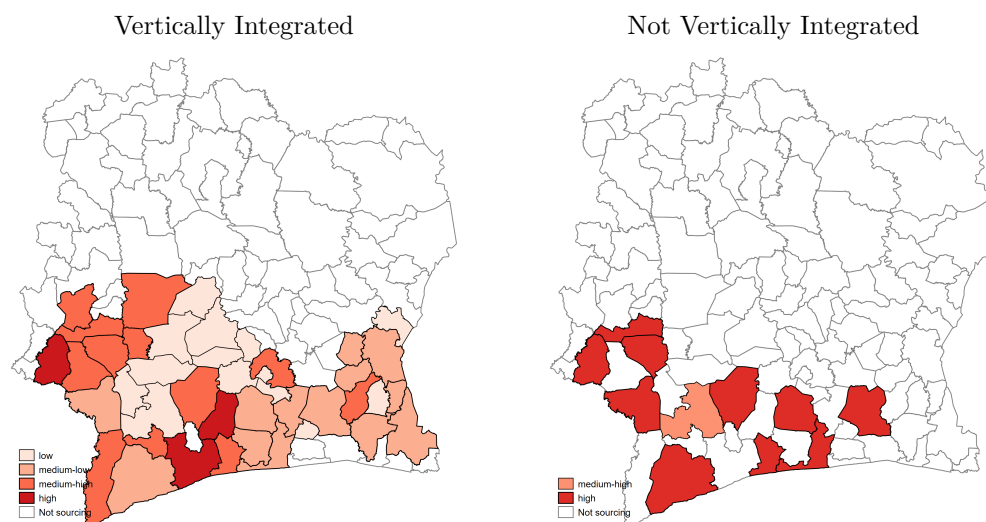


Figure A4: Cocoa deforestation in Côte d'Ivoire



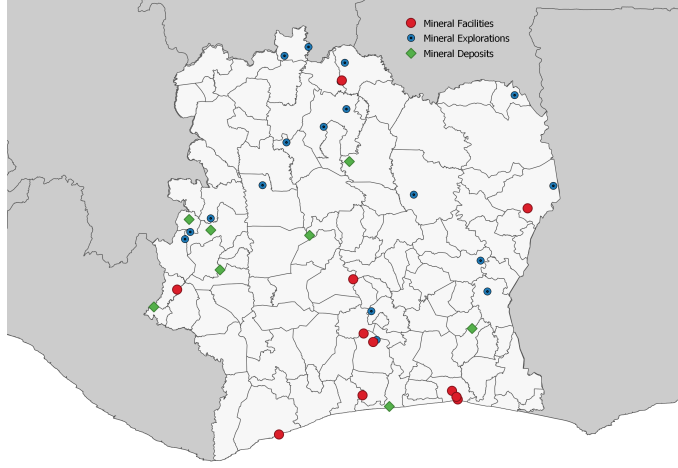
Notes: This figure shows cocoa deforestation (ha) in 2019. Calculated by identifying the total cocoa area in the target year that overlaps with the loss in undisturbed tropical moist forests for the 15-year period that begins 19 years before the target year (i.e., allocation period: 15 years, time to first harvest: 4 years). Source: Trase.

Figure A5: Importer Deforestation Exposure by Department



Notes: This figure shows the deforestation exposure for the importer Barry Callebaut by department in the year 2019 sourcing from integrated and non-integrated exporters (ETC).

Figure A6: Mineral activities



Notes: Location of Mining activities in Côte d'Ivoire in 2021. Source: U.S. Geological Survey (Padilla et al., 2021)

A.3 Importer Relationality

The economic rationale for relational sourcing rests on the ability of repeated interaction to sustain implicit agreements that cannot be fully specified in formal contracts (Baker et al., 2002; Macchiavello and Morjaria, 2015; Macchiavello, 2022). When a buyer and a supplier have interacted repeatedly and expect to continue doing so, the continuation value of the relationship raises the cost to the supplier of reneging on implicit commitments. In contexts where key transaction attributes, such as the traceability or environmental compliance of an agricultural commodity, are hard to verify or contract upon, this continuation value can substitute for formal enforcement.

In our theory framework, importer relationality serves as an empirical proxy for the continuation value of non-integrated links. A more relational importer maintains a concentrated network of core exporters. From any given exporter's perspective, a highly relational importer represents a more valuable long-term relationship to preserve. This can strengthen the credibility of implicit commitments to deliver low-risk cocoa on the non-integrated link and thereby narrow the deforestation gap between integrated and non-integrated links.

A growing body of empirical work on firm-to-firm trade distinguishes between ‘spot’ buyers spread, who short-term orders among many low-bidding sellers and ‘relational’ buyers, who concentrate orders with a small set of trusted exporters (Cajal-Grossi et al., 2023).

We follow the methodology of Cajal-Grossi et al. (2023) and Heise et al. (2024). For each importer i , product j , and origin country c , define the cell-level relationality as

$$Relational_{ijc} = -\frac{N_{ijc}^e}{N_{ijc}^s}, \quad (A1)$$

where N_{ijc}^s is the number of shipments and N_{ijc}^e is the number of distinct non-integrated

exporters in the importer–product–origin cell. The (negative) ratio of exporters to shipments is close to zero when many shipments are handled by few exporters, i.e., when sourcing is concentrated and relational, and more negative when purchases are dispersed across many exporters relative to shipments. The importer-level measure aggregates across cells using import values as weights:

$$Relational_i = \sum_{jc} \left[\frac{PQ_{ijc}}{PQ_i} \times Relational_{ijc} \right], \quad (A2)$$

where PQ_{ijc} is the import value for importer i in product–origin cell (j, c) and PQ_i is the importer’s total import value. The metric is then standardized. A higher (less negative) value of $Relational_i$ indicates more relational sourcing behavior.

For our baseline, we construct the metric using 2018 customs data from Côte d’Ivoire, Ecuador, and Ghana, which together account for over 70% of global cocoa production. Using multiple origin countries rather than a single country captures importers’ global sourcing strategies rather than strategies that might reflect country-specific institutions, and provides greater cross-importer variation. The metric is predetermined with respect to the deforestation outcomes measured from 2019 onward, which mitigates reverse-causality concerns.

Because importer–year fixed effects absorb the level of relationality, identification in (3) comes solely from the interaction with the vertical-integration indicator. A positive coefficient $\hat{\gamma} > 0$ on $VI_{ie} \times Relational_i$ indicates that the deforestation gap between integrated and non-integrated links is narrower for importers that rely more heavily on relational sourcing, consistent with the prediction that stronger continuation value on non-integrated links attenuates the integrated-link advantage.